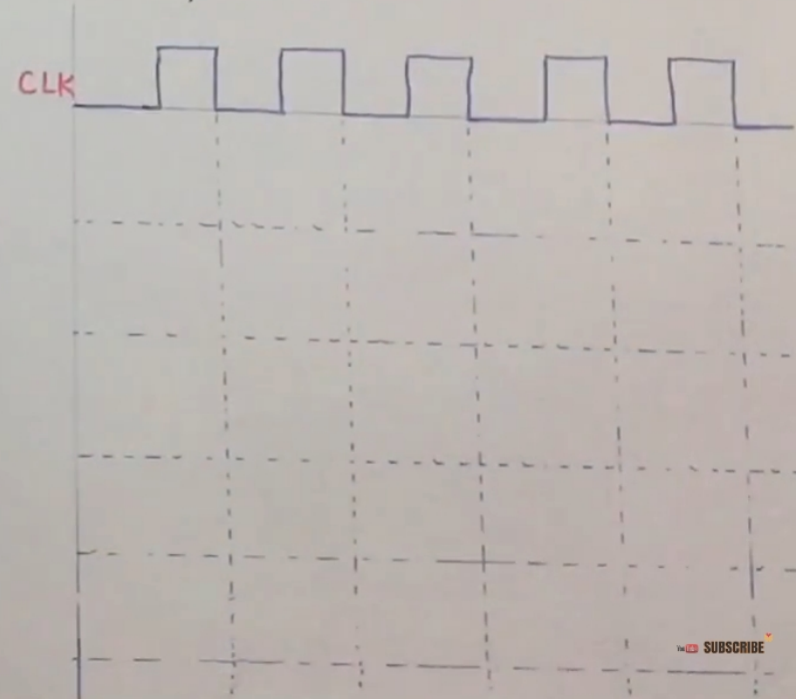


Ring Counter or Shift Register Counter

- Ring counter is a typical application of Shift register.
- The o/p of the Last FF is connected to the I/P of First FF.

Cr/P _r	CLK	Q ₀	Q ₁	Q ₂	Q ₃



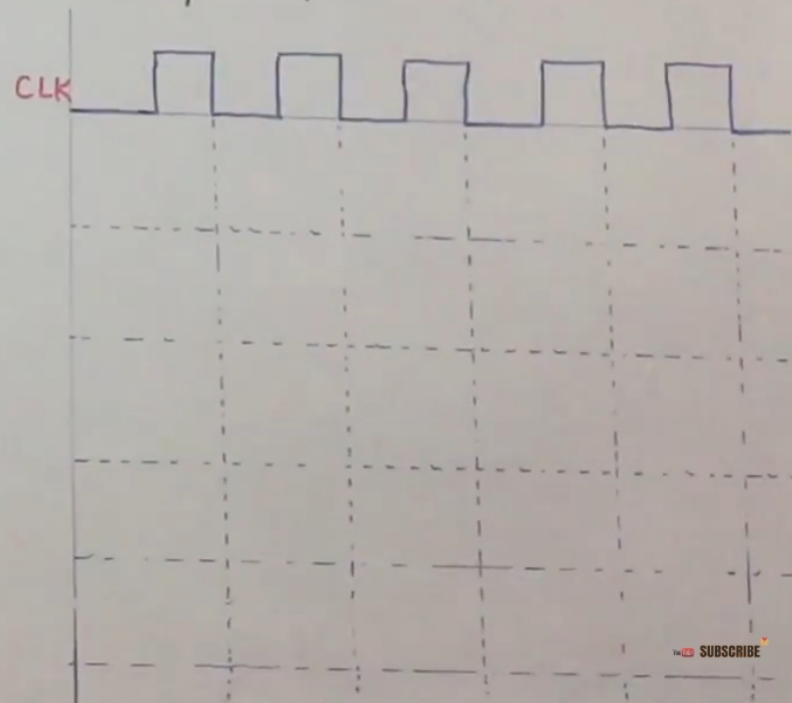
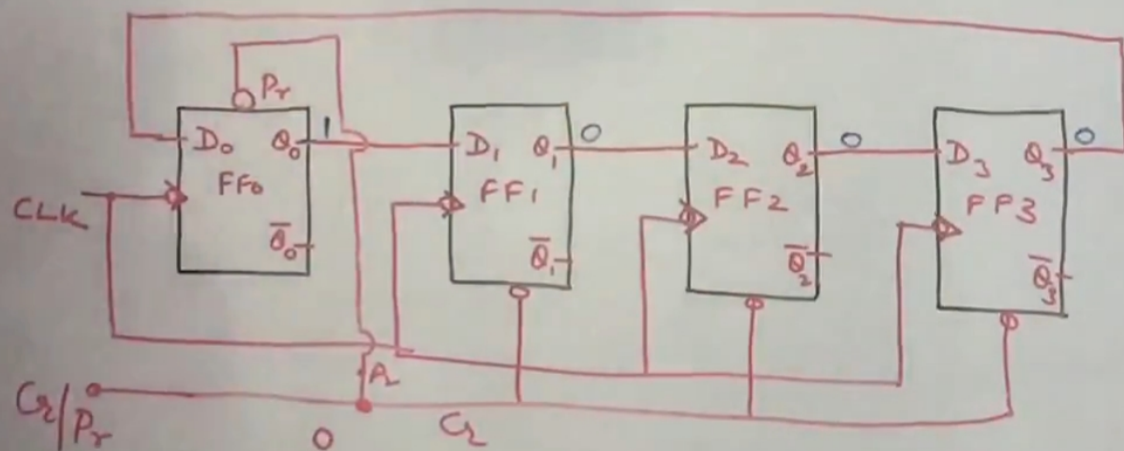
Ring Counter or Shift Register Counter

- Ring counter is a typical application of Shift register.

D	Q _{n+1}
0	0
1	1

$Cr=0$ $\rightarrow$ Active
 $Pr=0$ $\rightarrow$ FF $\rightarrow$ Q/P=1

Cr/Pr	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	1	0	0	0



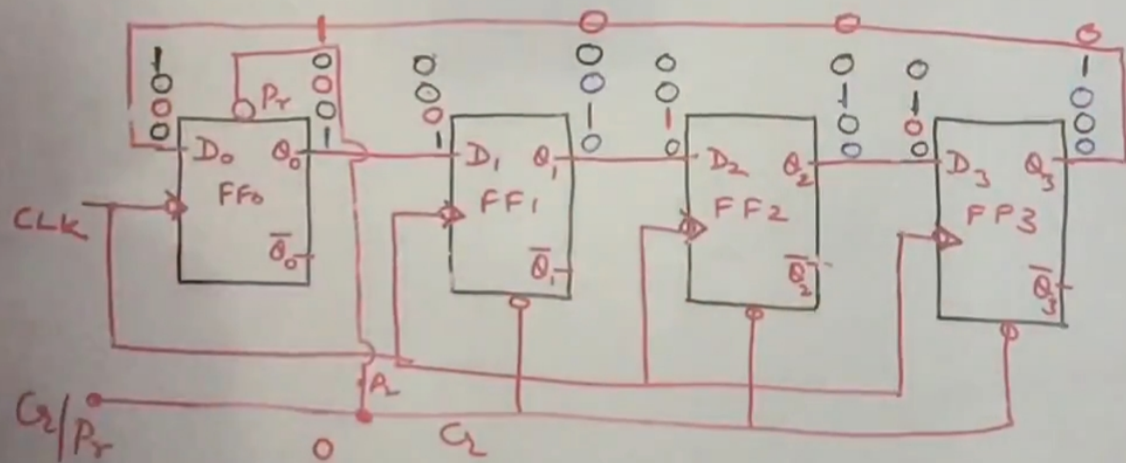
Ring Counter or Shift Register Counter

- Ring counter is a typical application of Shift registers.

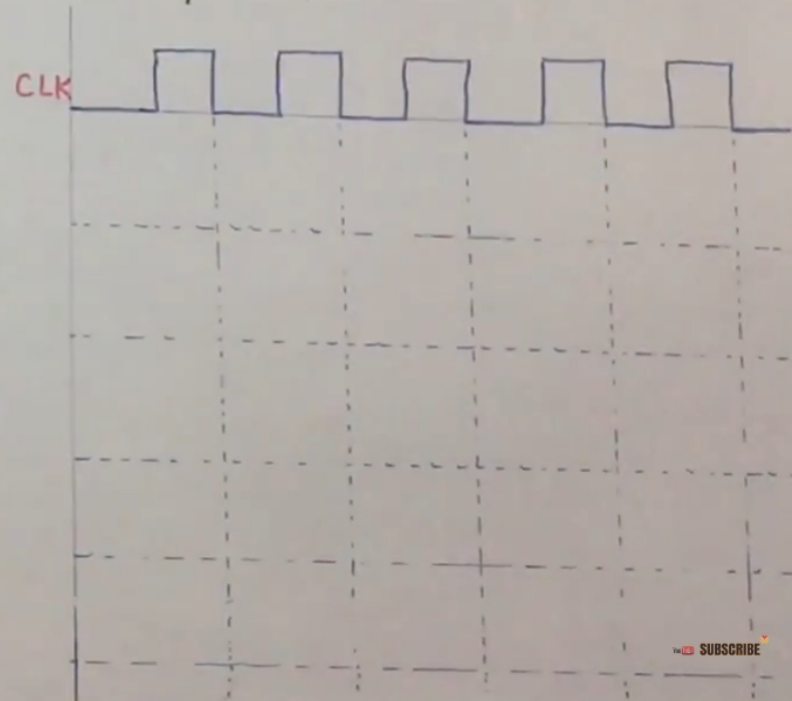
D	Q _{n+1}
0	0
1	1

$Cr=0 \rightarrow FF \rightarrow Q/P=0$
 $Pr=0 \rightarrow FF \rightarrow Q/P=1$
 Active

The o/p of the Last FF is connected to the I/P of First FF.



Cr/Pr	CLK	Q ₀	Q ₁	Q ₂	Q ₃
0	X	1	0	0	0
1	↓	0	1	0	0
1	↓	0	0	1	0
1	↓	0	0	0	1
1	↓	1	0	0	0



no. of States = No. of FF

27 States

